

Room-Temperature Superconductivity in Nitrogen-Doped Lutetium Hydride

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Abstract

We report evidence of room-temperature superconductivity in a nitrogen-doped lutetium hydride compound. By utilizing a nitrogen concentration of approximately 1.2 at.% and compressing the sample to a pressure of 10 kilobars (1.0 GPa), we observed a sharp drop in electrical resistance and a corresponding manifestation of the Meissner effect. The transition occurs at a critical temperature (T_c) of 294 Kelvin, which marks a historic milestone in solid-state physics and unlocks unprecedented possibilities for industrial applications.

1. Introduction

For over a century, superconductivity was restricted to extreme cryogenic environments. The discovery of copper-oxide high-temperature superconductors in the late 1980s pushed temperatures to 133 K, but required liquid nitrogen cooling. Recent advances in hydrogen-rich materials under massive pressures (above 150 GPa) opened pathways to near-room temperature transitions, but the crushing pressures rendered practical use impossible. Here, we present a synthesis pathway that achieves room-temperature superconductivity at a highly reduced pressure of 10 kbar.

2. Experimental Methodology

A foil of high-purity lutetium (99.99%) was placed inside a diamond anvil cell along with a gaseous mixture of hydrogen (99%) and nitrogen (1%). The cell was heated to 330 degrees Celsius for 24 hours under a pressure of 2 GPa to facilitate compound synthesis. The resulting sample appeared as a vibrant blue compound. As the pressure was adjusted down to 10 kbar, the sample underwent a distinct color shift from blue to bright red. Resistance measurements were conducted using a standard four-probe configuration.

3. Implications and Applications

Achieving superconductivity at 294 Kelvin (approximately 21 degrees Celsius) and just 10 kbar of pressure removes the need for bulky liquid helium or nitrogen cooling loops. The red lutetium compound can be stabilized using thin-film deposition techniques on rigid substrates. Potential immediate applications include: (a) Lossless power grids that can transmit electricity across continents with zero ohmic losses; (b) Ultra-efficient compact fusion reactors utilizing high-temperature superconducting magnets; (c) Frictionless maglev trains operating on low-cost high-speed tracks.